Limits to Arbitrage in the Indian Equity Market

Progress Report

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1 Summary

Since the project document of September 2026, I have done five things.

1. **Checked and refined the plan.** I re-read the primary sources (the SEBI and NSE circulars and the core papers) and corrected several institutional details in the project document. I also widened the framing: the project studies the limits to arbitrage in Indian equities as a whole, and the F&O ban is one of several sources of variation, not the only one.
2. **Built the theoretical framework you suggested.** One model covers all the limits the project studies (costs, short-sale constraints, arbitrage capital, risk and position limits). The F&O ban is modelled as the rule actually works. The model changed parts of the empirical design. Section 3 gives a three-page summary and Appendix A gives the full version.
3. **Built the ban and position-limit dataset** for every F&O stock from February 2010 to October 2026, directly from NSE archive files, and checked it against the exchange's rules.
4. **Built the price data and the main mispricing measure** (the futures-cash basis) for 2015 to 2026.
5. **Wrote down the tests before running them.** No mispricing measure has yet been compared with ban status, the reforms or any other treatment. The hypotheses, designs and the results that would contradict them are fixed first (Section 5), so that the theory and the data cannot be adjusted to each other after the fact.

Section 6 sets out the plan from here and Section 7 lists the points on which I would value your guidance.

2 Refinements to the project document

2.1 Scope

The research question is unchanged: how strongly do limits to arbitrage bind in Indian equities, and how much mispricing do they explain? India offers a dated, observable shock to almost every kind of limit, which lets the project study them side by side.

Kind of limit	Indian source of variation
Short-sale constraints	Only F&O-segment stocks are eligible for short selling and securities lending, and the practical overnight short is the single-stock future. Entry into and exit from the F&O segment changes this permanently; the F&O ban removes it temporarily.
Trading costs	Securities transaction tax (STT) on futures: 0.01% from June 2013, 0.0125% from April 2023, 0.02% from October 2024 and 0.05% from April 2026.
Arbitrage capital and funding	Arbitrage mutual fund assets (large inflows after the April 2023 change in debt fund taxation), money market rates, and the peak margin rules phased in from December 2020 to September 2021.
Arbitrage risk	Basis volatility and idiosyncratic volatility.
Position limits	The market-wide position limit (MWPL), entity limits, the ban rule, the March 2020 halving of the MWPL for volatile stocks and the October 2025 change in the MWPL formula.

2.2 Corrections to institutional details

- *Short selling.* The project document said there is no genuine short selling in the cash market. More precisely, naked short selling is prohibited, short selling must be covered by delivery, and under SEBI's framework only securities in the F&O segment are eligible for short selling and stock lending. Retail traders can short within the day; overnight short positions require stock lending, which is thin.
- *MWPL before October 2025.* The document gave the MWPL as 20% of free float "or 30 times the prior month's volume". The volume leg comes from a 2004 SEBI circular and was replaced in 2006 (SEBI/DNPD/Cir-31/2006). Throughout our sample the MWPL was 20% of free float, apart from the 2020 halving.
- *The ban rule.* A stock enters the ban when market-wide open interest exceeds 95% of the MWPL at the end of a day, and leaves it when open interest falls to 80%. During the ban every participant may only reduce positions, rollovers into the next expiry are not allowed, and from October 2025 the test uses delta-adjusted (futures-equivalent) open interest (NSE Clearing circular 119/2025). Increases are penalised at 1% of the value of the increase, subject to a minimum of Rs 5,000 and a maximum of Rs 1 lakh per entity per stock per day.
- *The 2025 changes.* The document described the October and December 2025 changes as relaxations. In fact the new formula, the lower of 15% of free float and 65 times average daily delivery (measured in shares), with a floor of 10% of free float, lowered the MWPL of every F&O stock (Section 4). The intraday checks of the MWPL from November 2025 are a tightening. The effect on how often bans occur depends on each stock, because open interest is now also measured in delta terms.
- *Other points.* NSE sets the daily settlement price of a future at a theoretical cost-of-carry value when the contract does not trade in the last half hour, which matters for measuring the basis. From August 2026 the closing price of F&O stocks comes from a closing auction. I could not find a source for the statement that position limits have been studied in Korea and Taiwan, so I have dropped it.

2.3 Changes to the design

- *Identification of the ban effect.* The project document proposed to use changes in the MWPL as an instrument for ban status. The model shows why this does not work as stated: a change in the MWPL moves the chance of a future ban, and traders respond to that chance even when no ban occurs. MWPL changes therefore identify the effect of the whole position-limit regime. The effect of the ban itself is studied with designs built around the ban rule (Section 5).
- *Mispricing measure.* The basis is built from daily volume-weighted prices of the future and the stock, not from settlement prices, because of the theoretical settlement rule above.
- *Constraint index.* The index keeps its role as a stand-alone measure, but it is built only from frictions (costs, lending, risk, position headroom). The basis and the ban, which are the outcome and the treatment, are no longer inputs.
- *Cross-section.* Besides the cross-sectional test of return signals, the project uses entry into and exit from the F&O segment. These events change the short-sale constraint for about two thousand listed stocks, which gives far more power than the 200 or so F&O stocks alone.
- *Earnings drift.* The earnings surprise is measured from the announcement-day return, which needs only prices. Earnings per share would be added if a source is available through the library.
- *Put-call parity* is kept as a robustness check rather than a main measure, because many stock options trade thinly.

3 Theoretical framework: summary

3.1 Set-up

The *basis* is the gap between the futures price and its cost-of-carry value, $m = \ln F - \ln F^*$, with $F^* = (S - \mathrm{PV}(D))e^{r(T-t)}$. Outside traders trade the stock and the future. Their demands depend on prices and on a demand shifter, so that without arbitrage the basis equals u/Φ, where u is the relative demand pressure toward futures and Φ measures the depth of the two markets. An arbitrage sector holds a units of the cash-and-carry trade (long stock, short future). Market clearing gives

$$m = \frac{u - a}{\Phi}.$$

Arbitrageurs choose a to maximise $am - \frac{\gamma}{2}a^2 - c_+a^+ - c_-a^-$. Here $\gamma = \alpha\sigma^2$ reflects risk aversion per unit of capital and the risk of the basis before convergence, c_+ is the cost of a cash-and-carry position and c_- the cost of a reverse position, which needs stock lending. Each Indian limit enters as a parameter: costs (STT, fees, funding) through $c_\pm$, short-sale constraints through c_-, capital and risk through γ, and position limits through caps on a.

3.2 The no-arbitrage band

Arbitrage starts only when the basis leaves the band $[-c_-, c_+]$. Inside the band the basis moves one-for-one with demand; outside it arbitrage absorbs part of every shock. This gives four

implications.

- Costs set the edges of the band, so a change in STT should move the edges by about the change in cost.
- Capital and risk set how far the basis travels beyond an edge.
- The edges shrink as expiry approaches.
- Because reverse arbitrage is expensive in India, the band sits below zero. Earlier Indian studies find exactly this.

When arbitrageurs already hold positions, the range in which they do not trade is $[\gamma a_{t-1} - c_u,\ \gamma a_{t-1} + c_+]$, where c_u is the cost of unwinding. The thresholds therefore move with the size of existing arbitrage positions. This matters for the threshold models used in the estimation.

3.3 The F&O ban

In a ban every participant, not only arbitrageurs, may only reduce positions. Two consequences follow.

First, arbitrage becomes one-sided. Arbitrageurs cannot open new cash-and-carry positions, so a future that is too rich is no longer corrected. They can still unwind existing positions (sell the stock, buy back the future), so a future that is too cheap is still corrected, at a floor $\gamma a_t - c_u$ that rises with the remaining positions. The floor wears out as positions are unwound, and an expiry inside the ban removes positions that cannot be rolled over.

Second, demand for new futures exposure is blocked. Part of it moves to the cash market (for example, a trader who cannot buy the future buys the stock on margin). This reverses the sign of the response of the basis: in normal times new long demand raises the basis, in a ban it lowers it.

Together these imply that the effect of a ban on the *level* of the basis has no fixed sign. It is positive when the pressure comes from traders closing positions and negative when new speculative demand keeps arriving. The empirical test is therefore whether the sign follows the composition of trading, not whether bans "widen" mispricing.

3.4 Other implications

- *Trading in a ban.* Every trade closes a long and a short position, so the change in open interest should equal minus the volume, and futures trading should thin out. On days without trades NSE's theoretical settlement price makes a settlement-based basis zero by construction.
- *Anticipation.* If exit is costlier during a ban, arbitrageurs ask for a higher return before opening positions as utilisation approaches 95%. The band widens smoothly before a ban, so a regression discontinuity at 95% captures only part of the effect.
- *The MWPL as an instrument.* Because the MWPL affects the chance of a future ban, it affects the basis even on days without a ban. It is not a valid excluded instrument for ban status.
- *Earnings news.* In a ban, informed traders cannot short through futures, so bad news is absorbed more slowly and the drift after bad news should be larger. Good news can still be traded in the cash market.

- *F&O entry.* Following Miller (1977), when investors disagree and cannot short, prices reflect the optimists. The model gives the overpricing in closed form; it rises with disagreement and falls when shorting becomes possible. Entry into the F&O segment should therefore lower overpricing, more so for stocks with high disagreement.

All closed-form results were checked numerically, and a simulation of the dynamic version reproduces the one-sided correction during bans.

4 Data built so far

4.1 Sources

All data come from free NSE archive files, downloaded and parsed with scripts so that every step can be repeated.

File	Content	Coverage
Combined open interest (daily)	MWPL, open interest, delta-adjusted open interest (from October 2025), status for the next day	Feb 2010 to Oct 2026
Securities in ban (daily)	Official ban list	Feb 2010 to Oct 2026
F&O bhavcopy (daily)	Futures prices, traded value, contracts, open interest; options activity by stock	2015 to 2026
Cash bhavcopy and delivery file (daily)	Prices, traded value and quantity, delivery quantity	2015 to 2026
Corporate actions	Dividends, splits and bonus issues with ex-dates	2014 to 2026
Interest rates	Three-month interbank rate (monthly)	2011 to 2026
Results filings	Release date and time of quarterly results, with links to the XBRL filings	2014 to 2026
Index values (daily)	Nifty 50 close and dividend yield	2015 to 2026

4.2 Checks

Before using the data I checked them against the exchange's own rules and against each other.

- The daily "no fresh positions" flags agree with the next day's official ban list in 99.85% of cases. Each remaining difference traces to a dated NSE file that carries another day's data, to a corrupt NSE file or to a special session.
- The ban rule holds in the data: utilisation was above 95% on the day before 99.2% of ban spells began, and at or below 80% on the last day of 99.5% of spells.
- The set of stocks in the open interest files matches the set of stock futures in the F&O bhavcopy exactly.
- Some NSE files needed care: about a hundred files write the year in two digits, a few carry another day's data under their own date, and some zip files bundle several days. The parsing handles all of these, and every correction is logged.
- Known gaps at the source: 13 corrupt open interest files in May and June 2012 (before the main sample) and the F&O bhavcopy of 30 March 2021.

4.3 First facts on bans

Table 1 summarises the bans from 2015. Over 2015 to 2026 there were 1,789 ban spells covering 9,414 stock-days and 141 stocks. Bans were most frequent in 2017 and 2024. A typical spell lasted four trading days up to September 2025. Under the new rules spells are fewer but longer: the 44 spells since October 2025 have a median length of nine days.

Table 1: F&O bans by year, 2015 to 2026

Year	F&O stocks (avg per day)	Stocks banned	Ban spells	Ban stock-days	Median spell (days)	Longest spell (days)
2015	154	16	69	367	4	21
2016	172	27	136	645	4	17
2017	201	43	287	1586	5	24
2018	206	38	162	810	4	31
2019	176	23	86	508	4	17
2020	140	40	178	756	3	13
2021	163	26	162	789	4	16
2022	196	23	100	457	4	13
2023	187	41	193	942	4	17
2024	185	49	289	1656	5	22
2025	216	37	94	556	5	15
2026[a]	209	10	33	342	10	29
2015 to 2026	183	141	1789	9414	4	31

[a] January to 27 October 2026. Spells are counted in the year they start.

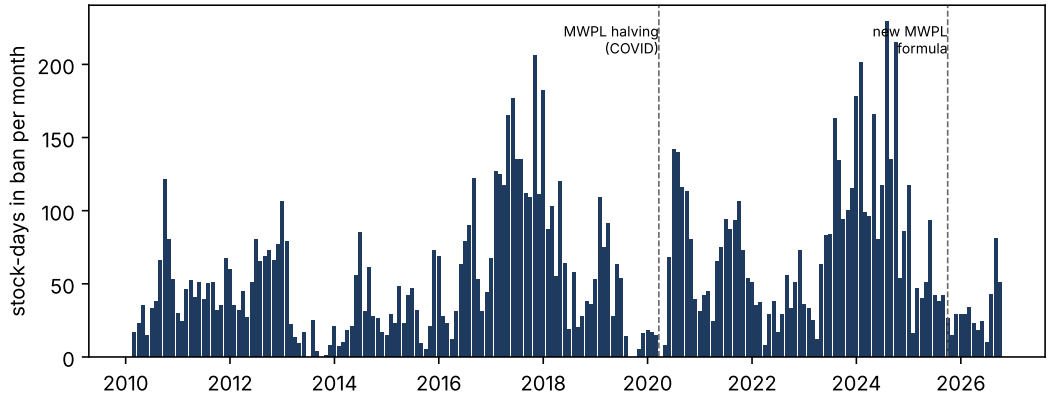


Figure 1: Stock-days in the F&O ban per month, 2010 to 2026.

4.4 Changes in the MWPL

- *March 2020.* The MWPL was halved for 61 stocks between 20 and 27 March 2020 and restored for 70 stocks on 26 November 2020. These dates match the regulatory record.
- *October 2025.* On 1 October 2025 the new formula lowered the MWPL of all 208 F&O stocks (Figure 2). The median stock's limit fell to 73% of its old value. 17% of stocks are at the floor (half the old limit) and 48% at 15% of free float (three quarters of the old limit). Because the size of the cut differs across stocks and was set by data from before the reform, it can be used as a measure of treatment intensity.

- *Regular updates.* Outside these events the MWPL changes mainly in February, May, August and November, when free float is updated from quarterly shareholding filings.

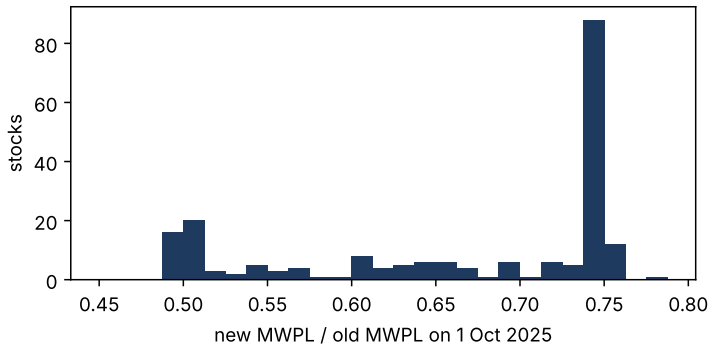


Figure 2: Ratio of the new to the old MWPL on 1 October 2025, all F&O stocks.

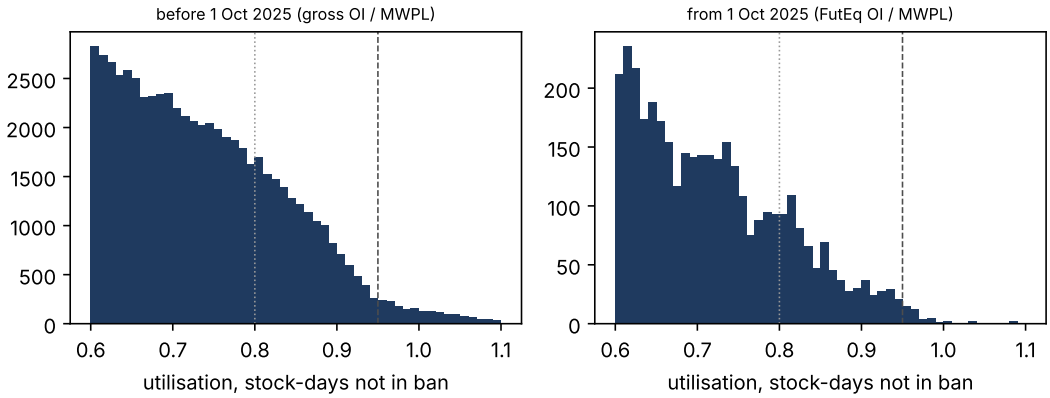


Figure 3: Utilisation of the MWPL on days a stock is not in a ban. Dashed line: 95% entry threshold; dotted line: 80% exit level.

Figure 3 shows how utilisation is distributed near the 95% threshold. Before the reform the density thins out smoothly through 95%, with no visible pile-up just below it. A formal density test comes before any regression discontinuity analysis.

4.5 F&O entry and exit, free float and results dates

- *F&O membership.* A stock is counted as in the F&O segment on a day if stock futures on it are listed that day. From 2015 to 2026 there were 239 entries and 171 exits; 162 of the exits are eligibility exits after which the stock kept trading in the cash market. Symbol changes (for example Zomato to Eternal) are linked through the ISIN so that they are not counted as an exit and an entry. The largest batches match the regulatory record: 45 entries on 29 November 2024 after the August 2024 revision of the criteria, and 34 exits on 28 June 2019 under the 2018 framework. An independent daily membership file for 2016 to 2025 agrees with this definition on every stock-day.
- *Free float.* Before October 2025 the MWPL was 20% of free float, so the free float of every F&O stock follows directly from the MWPL. Rebuilding the October 2025 limits from

pre-reform data reproduces NSE's figures exactly for all 85 stocks whose new limit is 15% of free float, which confirms both the formula and the free-float figures.

- *Results dates.* The release time of each quarterly result is known to the second. 63% of results are released after the market closes, so the event day is the next trading day in those cases. There are 7,910 results announcements for stocks while they are in the F&O segment, about four per stock per year.

4.6 The mispricing measure

For each stock and day the basis is computed from the volume-weighted average prices of the near-month future and of the stock, net of dividends going ex before expiry and of the cost of carry at the three-month interbank rate. The older F&O files do not report lot sizes, so lot sizes are recovered from open interest and checked against the day's price range: the futures average price lies within the day's high and low for 99.8% of traded contracts. Every futures observation has a matching cash price, and the trading days in the price files match those in the open interest files exactly.

Two checks led to changes. First, some corporate events other than dividends (for example the bonus debentures issued by NTPC in March 2015) move the futures price relative to the stock. Days on which such an event, or a dividend, goes ex before expiry are flagged and kept out of the main sample. Second, adjusting for dividends with their actual ex-dates over-corrects a little, probably because some dividends were not yet announced on the earlier days; these days are also flagged.

On the remaining 499,682 stock-days with at least five contracts traded, the median basis is 1.2 basis points, and the 10th and 90th percentiles are -33.0 and 26.1 basis points. The basis shrinks toward expiry, and its distribution is stable across the change in NSE's file format in July 2024. About 0.7% of stock-days lie more than 200 basis points from zero. These come mainly from stocks under heavy selling pressure (for example during business or credit distress), not from data errors, so they are kept and handled with the estimation methods. As a check on the cost-of-carry inputs, I compared the financing rate implied by Nifty futures with the three-month interbank rate, using the index's published dividend yield and contracts expiring between November and March, when few index dividends are paid. The two move together (monthly correlation 0.47) with an average gap of 0.7 percentage points, about the bias expected from using the full-year dividend yield in a low-dividend season. From 2021 the implied rate is higher, which suggests a premium in index futures over this period. As noted above, no mispricing measure has yet been related to bans, reforms or other treatments.

5 Pre-specified tests

The main hypotheses, how each will be tested and what result would contradict it:

Hypothesis	Design	Contradicted if
Bans make correction one-sided	Ban-entry event study against stocks that came close to 95% but did not cross; threshold models with ban-dependent adjustment	Positive and negative deviations change in the same way, or not at all
Bans move trading out of futures	Change in open interest against volume; futures and cash volume, delivery share	No change in futures activity
Costs set the band edges	Event windows around the STT changes of 2023, 2024 and 2026	Edges unrelated to the size of the cost change
Short-sale access lowers overpricing	Staggered difference-in-differences around F&O entry and exit, with matched controls	No change or the opposite sign
Position-limit regime	Continuous treatment intensity from the October 2025 formula; quarterly MWPL resets; behaviour as utilisation approaches 95%	No gradient before 95%, or no effect of the predicted change in utilisation
Level effect of bans	Ban-entry event study split by the composition of trading before the ban	Sign unrelated to the composition

Secondary hypotheses cover the constraint index, earnings drift in bans, arbitrage capital and return signals. Inference uses standard errors clustered by stock and date, with a correction for testing several main hypotheses. Power is an issue for the post-reform period, which has only 44 ban spells; results for that period will be reported with this caveat.

6 Plan from here

The steps below are in order of dependence.

1. Complete the remaining inputs: arbitrage fund assets and earnings per share from the XBRL filings.
2. Fix the test plan in writing, including sample filters, variable definitions and specifications, before any outcome is related to a treatment.
3. Estimate the no-arbitrage band with three-regime threshold error-correction models and smooth-transition models, and relate the band to the dated cost and capital changes.
4. Estimate the effects of the ban: event study, hysteresis comparison between 80% and 95% utilisation, regression discontinuity at 95% (as supporting evidence) and the October 2025 reform.
5. Estimate the effects of F&O entry and exit, and the cross-sectional test of return signals.
6. Build and validate the constraint index.
7. Write the policy chapter and the thesis, and release the code and the derived measures so the work can be reproduced.

7 Points on which I would value your guidance

1. Is the wider framing (limits to arbitrage in general, with the ban as one source of variation) acceptable to you?

2. Is the theoretical framework at the right depth for the thesis, or should parts of it be simplified or extended?
3. For the band estimation, would you prefer the three-regime threshold models, the smooth-transition models, or both? I would also welcome your advice on threshold models in which the thresholds depend on an observed state (existing arbitrage positions, or whether a stock is in a ban).
4. Can the department or library give access to a database with earnings per share and shareholding history (for example, CMIE Prowess)?
5. Would you like the test plan sent to you before I run the main estimates?

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A Theoretical framework: full version

The following pages give the full model with proofs, the numerical checks and the mapping from results to hypotheses.

Constrained Arbitrage under India's Limits: A Model of the Basis, the F&O Ban and Short-Sale Constraints

Theoretical framework for the BS Project

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Abstract

This note sets out one model that covers the limits to arbitrage the project studies in Indian equities: trading costs, the cost of shorting, arbitrage capital, arbitrage risk, position limits and the F&O ban. Each limit is a parameter of the same model, so every empirical hypothesis is derived from one framework. The core is a mean–variance arbitrage sector that trades the futures–cash basis against outside traders in both markets. Three features matter for the Indian setting. (i) The F&O ban is modelled as it actually operates: a *reduce-only* rule for every participant, not a freeze on the arbitrageur alone. (ii) Arbitrage inventory is a state variable, so the no-trade region moves with existing positions. (iii) Demand that is blocked from the futures market can spill into the cash market. The main prediction is that the ban turns two-sided arbitrage into *one-sided* arbitrage: futures that are too cheap are still corrected (by unwinding existing cash-and-carry positions), while futures that are too rich are not. The ban's effect on the *level* of the basis is *ambiguous*, and the model states the observable conditions that sign it. It also implies that the MWPL denominator is not a valid excluded instrument for ban status, because the same variable moves the probability of a future ban. All closed forms are checked numerically (Appendix A). The note ends with a map from propositions to hypotheses, each with the result that would contradict it.

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1 Purpose

The aim is to fix a theoretical framework before the empirical work, so that the hypotheses are derived rather than fitted. The model builds on the limits-to-arbitrage literature (Shleifer and Vishny, 1997; Gromb and Vayanos, 2010) and on the econometrics of no-arbitrage bands (Dwyer, Locke and Yu, 1996; Tse, 2001; Ters and Urban, 2020). Three design rules guide it:

1. **Institutions as they are.** The ban, the short-selling framework, settlement prices and costs are modelled from the primary sources (SEBI circular 2025/79; NSE Clearing circular 119/2025 and its FAQ of October 2025).
2. **Predictions that can fail.** Wherever the sign of an effect depends on the environment, the model says on what, and the empirical work tests that condition rather than a one-sided sign.
3. **Daily horizon.** The data are daily, so the relevant arbitrageur carries positions overnight (arbitrage mutual funds, proprietary desks), not a one-second market maker.

2 Environment

2.1 Prices and the basis

A stock with spot price S_t has a single-stock future maturing at T. With benchmark rate r_b (e.g. MIBOR) and known dividends, the cost-of-carry value is $F_t^* = (S_t - \mathrm{PV}_t(D))\, e^{r_b(T-t)}$. The *basis* (mispricing) is

$$m_t \;=\; \ln F_t - \ln F_t^*, \tag{1}$$

measured in basis points of the spot price. Write s and f for the deviations of spot and futures prices from the fundamental value, so $m = f - s$ to first order.

2.2 Outside traders and market clearing

Outside (non-arbitrage) traders in each market have linear demand. In the cash market net demand is $D_S - \psi_S s$, in the futures market net long demand is $D_F - \psi_F f$; $\psi_S, \psi_F > 0$ measure liquidity (the depth of the outside side of each market) and D_S, D_F are demand shifters. An arbitrage sector holds a units of the cash-and-carry trade: long a shares and short a futures ($a < 0$ is reverse arbitrage). Both markets are in zero net supply after removing the fundamental holdings, so

$$D_S - \psi_S s + a = 0, \qquad\qquad D_F - \psi_F f - a = 0. \tag{2}$$

Subtracting, the basis satisfies

$$m = \frac{u - a}{\Phi}, \qquad \frac{1}{\Phi} \equiv \frac{1}{\psi_F} + \frac{1}{\psi_S}, \qquad u \equiv \Phi\Big(\frac{D_F}{\psi_F} - \frac{D_S}{\psi_S}\Big). \tag{3}$$

u is the *relative demand pressure* toward futures: positive when outside traders want long exposure in futures more than in cash (e.g. leveraged speculative longs), negative when they want

short exposure in futures (hedgers, informed pessimists who can short only through futures). Without arbitrage the basis is u/Φ; arbitrage lowers it by a/Φ (buying cash raises s, selling futures lowers f).

2.3 The arbitrage sector

Assumption 1 (Arbitrage)**.** The arbitrage sector is competitive and chooses a to maximise

$$\Pi(a) = a\,m - \tfrac{\gamma}{2}a^2 - c_+\,a^+ - c_-\,a^-, \qquad \gamma \equiv \alpha\sigma^2 > 0, \tag{4}$$

where $a^{\pm} = \max(\pm a, 0)$, α is risk aversion per unit of capital (inversely related to arbitrage capital), σ^2 the basis risk borne until convergence (margin calls, early unwinds), c_+ the cost of a cash-and-carry position and c_- the cost of a reverse position.

Holding a unit of cash-and-carry to expiry earns m (the basis converges to zero), less costs. Decomposing,

$$c_+ = \tau_+ + (r_A - r_b)(T-t), \qquad c_- = \tau_- + (\ell_s + r_b - r_L)(T-t), \tag{5}$$

with $\tau_\pm$ the round-trip transaction costs of each trade (STT, stamp duty, exchange fees, brokerage, impact), r_A the required return of cash-and-carry capital, ℓ_s the stock-lending fee and r_L the rebate on short proceeds. Section 11 maps each Indian limit to these parameters.

2.4 Indian limits as parameters

Limit	Indian institution (dated shocks)	Parameter
Trading costs	STT on futures (sell side): 0.01% from 1 Jun 2013; 0.0125% from 1 Apr 2023; 0.02% from 1 Oct 2024; 0.05% from 1 Apr 2026. Cash delivery STT 0.1% each side; 0.1% on physically settled derivatives	τ_+, τ_-
Short-sale cost	Naked shorting barred; only F&O-segment securities are eligible for short selling and SLB; SLB thin; the overnight short is the future	c_- via ℓ_s; $c_- = \infty$ for cash-only stocks
Arbitrage capital	Arbitrage-fund AUM (tax-advantaged since the 1 Apr 2023 debt-fund tax change); funding rates; peak-margin phase-in (Dec 2020–Sep 2021)	α (so γ), r_A
Arbitrage risk	Basis volatility, idiosyncratic volatility	σ^2 (so γ)
Position limits	Entity limits (10/20/30% of MWPL from 1 Oct 2025); MWPL; the F&O ban (95% entry, 80% exit)	caps on a; reduce-only rule (Sec. 5)
Market depth	Liquidity of outside traders in each market	ψ_S, ψ_F (so Φ)

3 The no-arbitrage band

Proposition 1 (Asymmetric band and passthrough)**.** *Under Assumption 1 and (3), the equilibrium is unique:*

$$m(u) = \begin{cases} \dfrac{\gamma u + c_+}{\gamma\Phi + 1}, & u > \Phi c_+, \\ \dfrac{u}{\Phi}, & -\Phi c_- \le u \le \Phi c_+, \\ \dfrac{\gamma u - c_-}{\gamma\Phi + 1}, & u < -\Phi c_-, \end{cases} \qquad a(u) = \begin{cases} \dfrac{u - \Phi c_+}{\gamma\Phi + 1}, & u > \Phi c_+, \\ 0, & \textit{inside}, \\ \dfrac{u + \Phi c_-}{\gamma\Phi + 1}, & u < -\Phi c_-. \end{cases} \tag{6}$$

The basis is left alone inside the band $[-c_-, c_+]$; outside it, arbitrage absorbs a share of each demand shock, so the passthrough falls from $1/\Phi$ to $\gamma/(\gamma\Phi + 1)$.

Proof. Π is strictly concave with kinks at 0. The best response is $a = (m - c_+)/\gamma$ if $m > c_+$, $a = (m + c_-)/\gamma$ if $m < -c_-$, and $a = 0$ otherwise. The map $a \mapsto \mathrm{BR}((u - a)/\Phi) - a$ is strictly decreasing, so the fixed point is unique. Substituting $m = (u - a)/\Phi$ into the interior best responses gives (6); e.g. for the upper branch $m\Phi = u - (m - c_+)/\gamma$, i.e. $m = (\gamma u + c_+)/(\gamma\Phi + 1)$, which equals c_+ at $u = \Phi c_+$ (continuity). □

Proposition 2 (Comparative statics)**.** *(i)* Costs set the edges. *The band edges are c_+ and $-c_-$; they do not depend on capital or risk. A change in transaction costs $\Delta\tau_\pm$ (e.g. an STT change) shifts the corresponding edge one-for-one.*

(ii) Capital and risk set the passthrough. *Outside the band, $\partial m/\partial u = \gamma/(\gamma\Phi + 1)$ rises with $\gamma = \alpha\sigma^2$: scarcer capital or riskier convergence lets the basis travel further beyond the edge.*

(iii) Maturity. *By (5) both edges are affine in remaining maturity $T - t$ and converge to the round-trip costs τ_+, $-\tau_-$ at expiry (for positions held to expiry).*

(iv) Location. *The band is centred at $(c_+ - c_-)/2$, below zero whenever shorting costs more than carrying ($c_- > c_+$). Its* upper *edge is itself negative when $(r_b - r_A)(T - t) > \tau_+$, i.e. when cash-and-carry capital accepts a return sufficiently below the benchmark rate.*

Proof. (i)–(iii) follow from (6) and (5). (iv): the centre is $(c_+ - c_-)/2$; the sign of c_+ follows from (5). □

Remark. Part (iv) matters because the Indian evidence shows a band centred below zero: Shankar, Sankar and Kiran Kumar (2015) estimate mean thresholds of −39 and −10 bp for 102 NSE stock futures (5-minute data, 2007–09), and the median basis is negative in Aggarwal (2015) and Nandan et al. (2014). The model reproduces this if reverse arbitrage is costly and cash-and-carry capital accepts a below-benchmark return. One candidate is the tax treatment of arbitrage funds, which are taxed as equity. If so, the inflows after the 1 April 2023 debt-fund tax change should have lowered r_A and with it the upper edge, which is a testable implication. It must be separated from the futures STT increase on the same date, which raises τ_+.

4 Inventory and the moving no-trade region

The ban acts on existing positions, so the model needs inventory. Time is discrete (days). The arbitrage sector enters day t with inventory $a_{t-1} \geq 0$ and pays proportional costs on changes: c_+ per unit added and c_u per unit unwound (selling cash, buying back futures). To keep the dynamic analysis transparent, reverse arbitrage is set aside ($a_t \geq 0$): the capacity of stock *holders* to sell cash and buy futures when futures are cheap is part of the cash-market depth ψ_S. Each day the sector chooses a_t to maximise

$$a_t m_t - \tfrac{\gamma}{2} a_t^2 - c_+(a_t - a_{t-1})^+ - c_u(a_{t-1} - a_t)^+. \tag{7}$$

Proposition 3 (Inventory-dependent no-trade region)**.** *Given a_{t-1}, the sector trades only when the basis leaves*

$$\big[\, \gamma a_{t-1} - c_u, \;\; \gamma a_{t-1} + c_+ \,\big]. \tag{8}$$

In equilibrium, with $m_t^0 \equiv (u_t - a_{t-1})/\Phi$:

(a) if $m_t^0 > \gamma a_{t-1} + c_+$ the sector adds: $m_t = (\gamma u_t + c_+)/(\gamma\Phi + 1)$;

(b) if $m_t^0 < \gamma a_{t-1} - c_u$ it unwinds: $a_t = \max\{0, (u_t + \Phi c_u)/(\gamma\Phi + 1)\}$ and $m_t = (u_t - a_t)/\Phi$;

(c) otherwise $a_t = a_{t-1}$ and $m_t = m_t^0$.

Proof. (7) is strictly concave with kinks at a_{t-1}. The marginal value of the position is $m_t - \gamma a_t$, so adding is optimal iff $m_t - \gamma a_{t-1} > c_+$ and unwinding iff $m_t - \gamma a_{t-1} < -c_u$; interior solutions are $a_t = (m_t - c_+)/\gamma$ and $a_t = (m_t + c_u)/\gamma$. Uniqueness and the closed forms follow as in Proposition 1, combining each branch with (3). □

Empirical content. Thresholds are not constants: they rise with arbitrage inventory. A large existing cash-and-carry book (proxied by open interest built while the basis was rich) raises both the level at which new arbitrage starts and the level at which unwinding starts. Threshold models should allow thresholds to depend on lagged OI.

5 The F&O ban as a reduce-only rule

5.1 The rule

Once market-wide open interest in a stock's derivatives exceeds 95% of its MWPL at the end of a day, the stock enters the ban from the next day. Every entity may then only trade to reduce its position (from 1 October 2025, its end-of-day futures-equivalent delta); a change of sign counts as a violation. Normal trading resumes when open interest falls to 80% of MWPL (NSE Clearing circular 119/2025). The rule covers all expiries and options; rolling a position into the next expiry is not permitted. It binds outside traders as much as arbitrageurs.

Assumption 2 (Ban)**.** During a ban:

(B1) *Reduce-only arbitrage:* $0 \leq a_t \leq a_{t-1}$.

(B2) *Blocked new exposure:* outside traders' demand shocks have two parts, $\Delta u_t = \varepsilon_t^n + \varepsilon_t^c$ in normal times. Closing demand ε_t^c (long liquidation, short covering) can still be executed and enters as usual. Demand for *new* futures exposure ε_t^n cannot; a share $\chi \in [0,1]$ of it moves to the cash market and the rest is withdrawn. In the units of (3) the ban maps ε_t^n into $-\lambda\varepsilon_t^n$, with $\lambda \equiv \chi\,\psi_F/\psi_S \geq 0$.

(B2) follows from (3): new long demand $\Delta D_F = \varepsilon$ raises u by $\Phi\varepsilon/\psi_F$ if it enters futures, but lowers u by $\Phi\chi\varepsilon/\psi_S$ if a share χ of it buys cash instead. A leveraged retail trader who cannot buy the future and buys the stock on margin is the leading case.

Lemma 1. *For the same u_t and a_{t-1}, the basis in a ban is at least as high as in normal times.*

Proof. (B1) removes only the option to add, which is used only when it lowers m (branch (a) of Proposition 3); branches (b)–(c) are unchanged. □

Proposition 4 (One-sided correction)**.** *Let a ban start with inventory $a_{t-1} > 0$.*

(i) No ceiling. *For $u_t \geq a_{t-1}(1+\gamma\Phi) - \Phi c_u$ the basis is $m_t = (u_t - a_{t-1})/\Phi$: every increase in pressure passes through fully ($\partial m/\partial u = 1/\Phi$ instead of $\gamma/(\gamma\Phi+1)$ beyond the normal upper edge). Rich futures are not corrected.*

(ii) A floor while inventory lasts. *For lower u_t the sector unwinds and $m_t = \gamma a_t - c_u$, so cheap futures are still corrected, at a level that* rises *with the remaining inventory.*

(iii) The floor wears out. *Once $a_t = 0$ the floor disappears and $m_t = u_t/\Phi$. Because positions cannot be rolled, an expiry inside a ban removes the expiring part of the inventory for the rest of the ban.*

Proof. Apply Proposition 3 with branch (a) deleted by (B1). Branch (c) holds while $m_t^0 \geq \gamma a_{t-1} - c_u$, i.e. $u_t \geq a_{t-1}(1+\gamma\Phi) - \Phi c_u$, giving (i). In branch (b), $a_t = (m_t + c_u)/\gamma$ when interior, so $m_t = \gamma a_t - c_u$, giving (ii). If $(u_t + \Phi c_u) \leq 0$ then $a_t = 0$ and $m_t = u_t/\Phi$, giving (iii). Expiry: delivered positions close and cannot be re-opened in the next expiry under the ban. □

Proposition 5 (Sign reversal of new-exposure demand)**.** *In normal times a shock to demand for new futures exposure moves the basis in its own direction: $\partial m_t/\partial\varepsilon_t^n = 1/\Phi$ inside the no-trade region and $\gamma/(\gamma\Phi+1)$ beyond its upper edge. In a ban the same shock moves it the other way: $\partial m_t/\partial\varepsilon_t^n = -\lambda/\Phi$ inside the no-trade region and $-\lambda\gamma/(\gamma\Phi+1)$ on the unwinding floor. The reversal is strict iff $\chi > 0$.*

Proof. Chain rule on Propositions 3–4, with $\partial u/\partial\varepsilon^n = 1$ in normal times and $-\lambda$ in a ban by (B2). □

Proposition 6 (The level effect is ambiguous)**.** *Fix a_{t-1}, the pre-shock pressure u^0 and closing demand ε^c, and compare the same new-exposure shock $\varepsilon^n \geq 0$ with and without a ban: $m^N = m(u^0 + \varepsilon^c + \varepsilon^n)$ and $m^B = m^{ban}(u^0 + \varepsilon^c - \lambda\varepsilon^n)$. Then:*

(i) $m^B - m^N$ is non-increasing in ε^n;

(ii) at $\varepsilon^n = 0$, $m^B - m^N \geq 0$, strictly if closing pressure would have triggered new arbitrage in normal times;

(iii) $m^B - m^N < 0$ *for* ε^n *large enough.*

The ban raises *the basis when pressure comes mainly from closing trades (e.g. short covering) and* lowers *it when new speculative demand keeps arriving; the observable sign condition is the composition of order flow during the ban.*

Proof. m^N is non-decreasing in ε^n; m^B is non-increasing in ε^n because u^B falls in ε^n and m^{ban} is non-decreasing in u. Hence (i). (ii) is Lemma 1. (iii): m^N rises without bound (slope at least $\gamma/(\gamma\Phi + 1) > 0$) while m^B is non-increasing. □

Proposition 7 (Trading and measurement in a ban)**.** *(i) If the reduce-only rule binds contract by contract, every futures trade in a ban closes a long and a short, so* $\Delta OI_t = -V_t$ *(volume) outside expiry days. Trades that offset across instruments (e.g. delta-reducing option trades allowed from 1 October 2025) break the identity; the gap measures such offsetting.*

(ii) Trading requires that some long and some short both want to reduce at the same price, so futures volume falls and quoted spreads widen relative to normal days.

(iii) On days without trades in the last half hour, NSE sets the futures settlement price at a theoretical cost-of-carry value. The settlement-price basis is then zero by construction (up to differences between NSE's rate and r_b*).*

Proof. (i) A buyer may only reduce a short and a seller only reduce a long, so each trade lowers OI by its size. (ii) follows from the restriction of the trading set. (iii) is the NSE settlement rule. □

Remark. Part (iii) is a trap for the empirics. Bans thin out trading, so a naive basis built from settlement prices would show the basis "closing" during bans for purely mechanical reasons. The data work must use trade-based prices and flag theoretical settlements.

6 Anticipation and hysteresis

Assumption 3 (Exit in a ban is costlier)**.** If a ban occurs before a position is closed, unwinding costs $c_u + \ell$ with $\ell > 0$, because the counterparties are only traders who also want to reduce (Prop. 7(ii)). The probability that a position opened when utilisation is $\omega = OI/MWPL$ meets a ban before it is closed is $\pi(\omega)$, continuous and increasing.

Proposition 8 (Anticipation)**.** *Outside a ban the effective cost of opening cash-and-carry is* $c_+ + \pi(\omega)\ell$*. The upper edge of the no-trade region rises smoothly as utilisation approaches 95%, and new arbitrage slows before any ban. The jump at* $\omega = 0.95$ *consists only of the reduce-only effects of Section 5; anticipation is continuous through the threshold.*

Proof. The expected cost of a unit opened at ω is $c_+ + \pi(\omega)\ell$ (exit at a later date is costlier with probability π); substitute for c_+ in Proposition 3. Continuity of π gives continuity at 0.95. □

Remark (What the density of utilisation should look like)*.* Arbitrageurs hold back as ω rises, which thins the density just below 95%; speculators who expect a ban may rush to open positions first, which does the opposite. The model therefore does not sign a McCrary test; the test tells us which force dominates.

Corollary 1 (Hysteresis)**.** *For $\omega \in [0.80, 0.95)$ a stock may be in a ban (it crossed 95% earlier) or not. At the same ω the first is governed by Section 5, the second by Proposition 8. Their difference is the effect of the reduce-only state relative to the anticipation state, which is the effect the 95/80 rule creates.*

7 What the model implies for identification

Proposition 9 (MWPL is not a clean instrument for ban status)**.** *The basis depends on the MWPL through three channels: the ban state $B_t = B(\omega_t, B_{t-1})$, the anticipation term $\pi(\omega_t)$, and entity limits proportional to the MWPL when they bind. A change in the MWPL therefore moves the basis even in periods with no ban, so it violates the exclusion restriction for an instrument of B_t. Denominator variation identifies the* total *effect of the position-limit regime (ban plus anticipation plus caps), not the effect of the ban alone.*

Proof. $\omega = OI/MWPL$ enters $\pi(\omega)$ (Prop. 8) independently of B. □

What each design identifies under the model:

1. *RD at 95%*: the local jump in the reduce-only effects, provided utilisation cannot be placed precisely on one side.
2. *Hysteresis comparison* (Cor. 1): the reduce-only state against the anticipation state at equal ω; selection on past paths must be controlled (lagged utilisation, lagged basis).
3. *Ban-entry event study with near-miss controls*: Props. 4–7 dynamically, against stocks with similar paths that stayed below 95%.
4. *1 October 2025 reform and quarterly MWPL recalculations*: reduced-form effects of predicted changes in utilisation (new formula; FutEq numerator) on the basis. These are total-regime effects (Prop. 9).

8 Short-sale constraints beyond the basis

8.1 Earnings news in a ban

Proposition 10 (Bad news is incorporated more slowly in a ban)**.** *Let an announcement reveal news $\pm\delta$, and let trading on the day impound a share k of it. For bad news, a share ρ of informed selling capacity runs through short futures, the only overnight short in an F&O stock. In a ban that capacity is unavailable, so the share impounded falls to $k(1-\rho)$ and the post-announcement drift after bad news rises from $(1-k)\delta$ to $(1-k+k\rho)\delta$. Good news is unaffected because optimists can buy cash.*

Proof. Immediate from the definitions; the asymmetry comes from the fact that the cash market allows buying but not overnight selling by non-holders. □

Remark. Outside a ban the same mechanism predicts that informed pessimists push the futures price *below* carry before bad news, since their short futures cannot be met by reverse arbitrage. A recent working paper (Saurav, Srivastava and Agarwalla, 2026) reports a negative basis before

earnings announcements in Indian stock futures. The model adds that this pre-announcement signal should *vanish* for stocks in a ban, while drift after bad news should grow.

8.2 Cash-only stocks and F&O inclusion

Stocks outside the F&O segment have no single-stock futures and are not eligible for short selling or SLB under SEBI's framework, so c_- is effectively infinite for overnight positions. A Miller (1977) mechanism then applies.

Proposition 11 (Overpricing without a short channel)**.** *A continuum of investors has valuations $v_i \sim U[v-\delta, v+\delta]$ and demands $k(v_i - p)$ if long; shorting costs c_s per unit, so investors with $v_i < p - c_s$ short $k(v_i - p + c_s)$. Supply is $Q > 0$.*

(i) The price $p(c_s)$ is non-decreasing in c_s.

(ii) With free shorting $p = v - Q/k$. With no shorting and $\delta \geq Q/k$, $p = v + \delta - 2\sqrt{\delta Q/k}$, so the overpricing is $\left(\sqrt{\delta} - \sqrt{Q/k}\right)^2$, increasing in dispersion δ.

Proof. (i) Aggregate demand falls in p and rises in c_s (short positions shrink), so the market-clearing price rises in c_s. (ii) Free shorting: $\int k(v_i - p)\, di = k(v-p) = Q$. No shorting: $\frac{k}{2\delta}\int_p^{v+\delta}(x-p)\,dx = \frac{k(v+\delta-p)^2}{4\delta} = Q$, valid when $p \geq v - \delta$, i.e. $\delta \geq Q/k$. Subtract; the derivative in δ is $1 - \sqrt{Q/(k\delta)} > 0$. □

Empirical content. F&O inclusion lowers c_s (futures, options, short-selling and SLB eligibility arrive together). It should lower prices of high-dispersion stocks at inclusion and reduce their subsequent underperformance; exclusion should do the reverse. Diamond and Verrecchia (1987) imply that, even without a level bias, short constraints slow the incorporation of bad news, which points in the same direction as Proposition 10.

9 Heterogeneous arbitrageurs and the estimation lens

With a single cost c_+ the band has sharp edges and, under impulse-style correction, the outer regimes would almost never be observed. Let costs differ across arbitrageurs, $c \sim G$ on $[\underline{c}, \bar{c}]$, each with the objective (4). Aggregate arbitrage is $a(m) = \int (m-c)^+\, dG(c)/\gamma$, which is continuous and convex in m above $\underline{c}$, so the strength of correction rises smoothly with the size of the deviation. In discrete time, with capital moving to opportunities only gradually, the basis follows an error-correction process whose speed of adjustment is a smooth, increasing function of $|m_{t-1} - \text{centre}|$. This is the exponential or logistic smooth-transition model (Tse, 2001). It is also well approximated by a three-regime threshold model whose thresholds are quantiles of G (Dwyer, Locke and Yu, 1996; Ters and Urban, 2020). Proposition 3 adds that the thresholds move with inventory, and Proposition 4 that in a ban the upper regime's adjustment speed falls toward zero while the lower regime's remains. A band-of-inaction (impulse-control) derivation gives the same reading for a single arbitrageur: the threshold is the point at which correcting the deviation becomes worth its cost.

10 Calibration and horizon

For an overnight carrier, the round-trip cost of a cash-and-carry trade includes STT on the cash purchase and sale (0.1% each), STT on the futures sale (0.01–0.05% depending on the period) or 0.1% on physical delivery, stamp duty, exchange and clearing fees, brokerage and impact. That is tens of basis points, far wider than the bands of a few basis points estimated from one-second data (Aggarwal, 2015), which describe intraday market makers. Arbitrage funds that roll rather than deliver spread these costs over several expiries, which lowers the effective τ_+ per month. The empirical work will compute a bottom-up cost band for each period; the simulations here use round illustrative numbers ($c_+ = 15$, $c_u = 10$, $c_- = 60$ bp) and are not estimates.

11 From propositions to hypotheses

H	Source	Prediction	What would contradict it
H1	Props. 1–3	Mispricing size and persistence rise with ex-ante frictions	No relation between friction index and band/persistence
H2	Props. 10, 11	Drift after bad news larger without a short channel (cash-only stocks; ban days); good news unaffected	Symmetric drift response, or larger effect for good news
H3	Prop. 2(i),(iii)	Edges move one-for-one with cost changes (STT 2023, 2024, 2026); edges shrink toward expiry; estimated band close to cost band	Band unrelated to dated cost changes
H4	Prop. 2(ii),(iv)	Scarcer capital widens excursions beyond the edges; tax-advantaged inflows lower the upper edge	Basis level unaffected by arbitrage-fund flows (after separating the 1 Apr 2023 STT change)
H5	Prop. 11	Overpricing-side anomalies stronger where shorting is costly; underpricing side unaffected	Symmetric effect on both legs
H6	Prop. 11	F&O inclusion lowers overpricing, more for high-dispersion stocks; exclusion raises it	No change, or opposite sign
H7	Prop. 4	In a ban, positive deviations stop reverting; negative ones still revert while inventory lasts, less after an expiry in the ban	Symmetric change, or no change
H8	Props. 5, 7	In a ban, futures volume falls, $\Delta OI \approx -V$, cash activity rises; response of the basis to new-demand shocks flips sign	Futures activity unchanged; no sign flip
H9	Props. 8, 9, Cor. 1	Upper edge rises smoothly as utilisation nears 95%; the 1 Oct 2025 package acts through predicted utilisation, with sign set per stock	Jump only at 95% with no pre-trend; no effect of predicted utilisation

The level effect of the ban (Prop. 6) is deliberately *not* a one-sided hypothesis: the test is whether

its sign follows the composition of order flow (closing versus new demand) as the model says.

12 Limitations

1. Partial equilibrium with linear outside demand and a myopic, mean–variance arbitrage sector; dynamic hedging and the option to unwind early (Brennan and Schwartz, 1990) enter only through the inventory structure.
2. The split of outside demand into closing and new-exposure parts (B2) is the modelling choice that drives Props. 5–6. Its empirical counterpart (order flow that opens versus closes positions) is only partly observable from public data (changes in OI against volume).
3. Reverse arbitrage by stock holders is folded into cash-market depth, not modelled as a separate agent.
4. Anticipation rests on Assumption 3 (costlier exit in a ban), which is itself testable (Prop. 7(ii)).
5. Propositions 10–11 are deliberately simple; they fix directions, not magnitudes.

A Numerical verification and illustration

A Python script solves each equilibrium by brute force (bisection on the best-response fixed point) and compares it with the closed forms. All checks pass: the static band (Prop. 1), the inventory step in normal times and in a ban (Props. 3–4), the floor $\gamma a_t - c_u$ and the full passthrough above it, Lemma 1, the monotonicity and sign change in Proposition 6, and the Miller closed forms (Prop. 11).

The simulation uses illustrative parameters ($\Phi = \gamma = 1$; $c_+ = 15$, $c_u = 10$ bp; demand pressure AR(1) with persistence 0.9 around 25 bp; daily shock s.d. 6 bp for each component; $\lambda = 0.5$). Each experiment runs the same shocks with and without a 40-day ban.

- *Asymmetric reversion* (300 episodes): the slope of Δm_t on m_{t-1} above the 80th percentile weakens from -0.18 (no ban) to -0.13 (ban); below the 20th percentile it strengthens from -0.12 to -0.16.
- *Level effect*: with no extra new-exposure demand during the ban the mean basis is $+4.9$ bp higher; with a steady inflow of 1, 2 and 4 bp/day of new demand it is -3.1, -10.8 and -26.0 bp lower (Prop. 6).

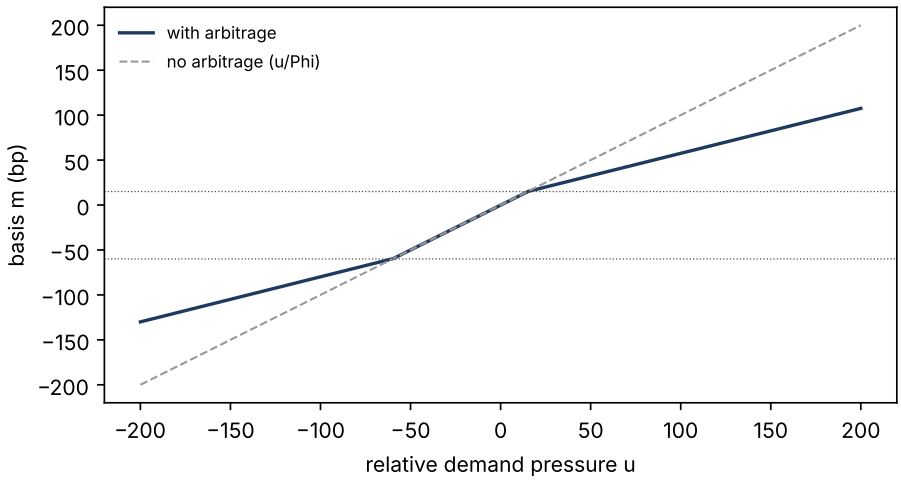


Figure 1: Static equilibrium (Prop. 1): the basis follows demand pressure one-for-one inside the band and is compressed outside it. Dotted lines: edges c_+ and $-c_-$.

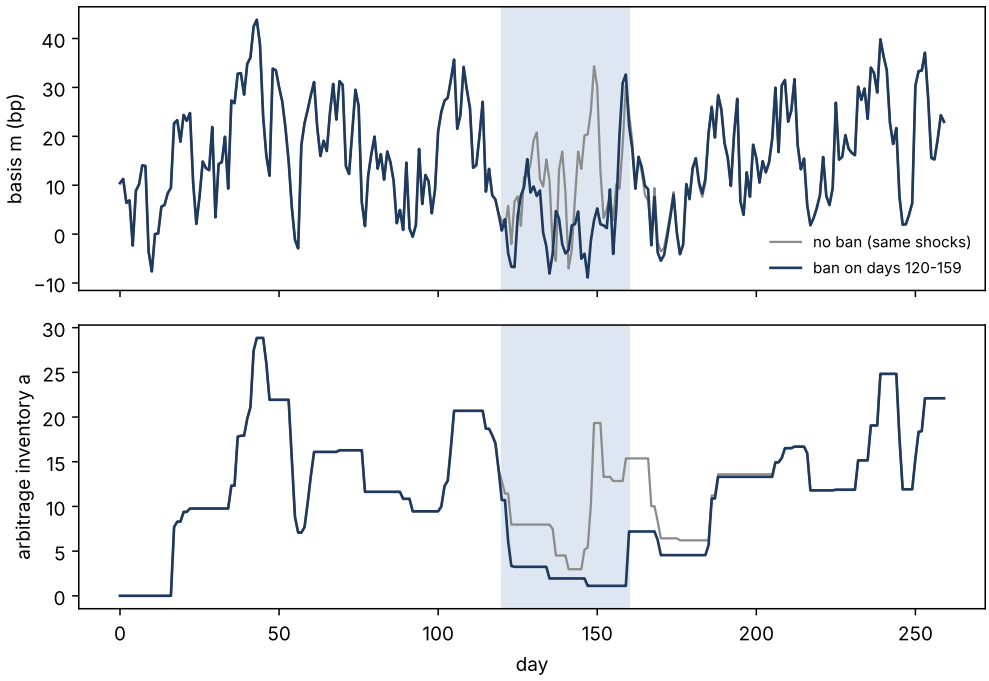


Figure 2: One simulated path, same shocks with and without a ban on days 120–159. In the ban, inventory can only fall and upward pressure passes through fully.

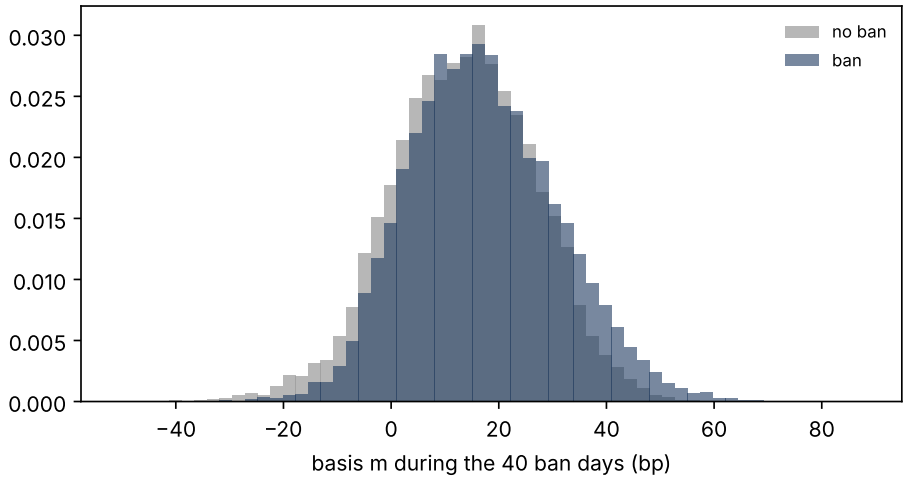


Figure 3: Distribution of the basis over the 40 ban days, pooled across 300 paired simulations.

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